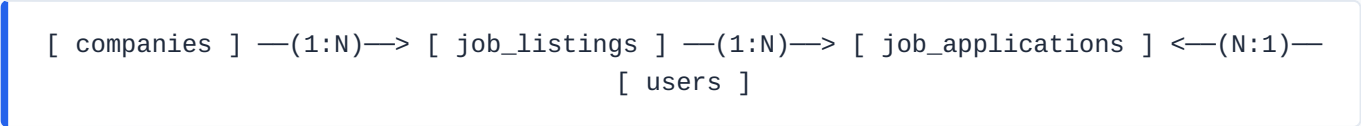


Database Schema Specification

SQLAlchemy ORM Model Architecture & Entity Specifications

1. System Overview & Architecture

The database schema defines a relational structure for a job application portal, comprising 4 primary entities managing users, companies, job postings, and application tracking.



2. Entity & Table Specifications

users

Stores applicant profile and account metadata.

COLUMN	TYPE	NULLABLE	DEFAULT	CONSTRAINTS	DESCRIPTION
id	INTEGER	No	Auto-inc	PRIMARY KEY	Unique user identifier.
name	VARCHAR(100)	No	—	—	Full name of the applicant.
email	VARCHAR(254)	No	—	UNIQUE	Primary email address.
created_at	DATETIME	No	datetime.now	—	Account creation timestamp.

ORM Relationships:

• job_applications:

One-to-Many (1:N) relationship to JobApplication. Cascades all, delete-orphan.

companies

Stores organization details posting job openings.

COLUMN	TYPE	NULLABLE	DEFAULT	CONSTRAINTS	DESCRIPTION
id	INTEGER	No	Auto-inc	PRIMARY KEY	Unique company identifier.
name	VARCHAR(150)	No	—	—	Official company name.
industry	VARCHAR(100)	No	—	—	Industry domain (e.g., Software).
location	VARCHAR(150)	No	—	—	Headquarters location.
website	VARCHAR(255)	Yes	NULL	—	Company website URL.
created_at	DATETIME	No	datetime.now	—	Record creation timestamp.

ORM Relationships:

• job_listings:

One-to-Many (1:N) relationship to JobListing. Cascades all, delete-orphan.

job_listings

Stores individual job postings managed by companies.

COLUMN	TYPE	NULLABLE	DEFAULT	CONSTRAINTS	DESCRIPTION
id	INTEGER	No	Auto-inc	PRIMARY KEY	Unique listing identifier.
company_id	INTEGER	No	—	FK -> COMPANIES.ID	Parent company ID.
title	VARCHAR(200)	No	—	—	Job title.
description	TEXT	No	—	—	Detailed job description.
location	VARCHAR(150)	No	—	—	Job location / Remote status.
posted_date	DATE	No	—	—	Date job was published.
closing_date	DATE	Yes	NULL	—	Application deadline.
created_at	DATETIME	No	datetime.now	—	Record creation timestamp.

Foreign Key Behavior & Relationships:

- **Foreign Key Constraint:** `company_id` references `companies.id` with **ON DELETE CASCADE**.
- **company:** Many-to-One (**N:1**) relationship back to **Company**.
- **job_applications:** One-to-Many (**1:N**) relationship to **JobApplication**. Cascades **all**, **delete-orphan**.

job_applications

Junction entity linking applicants to specific job openings.

COLUMN	TYPE	NULLABLE	DEFAULT	CONSTRAINTS	DESCRIPTION
id	INTEGER	No	Auto-inc	PRIMARY KEY	Unique application identifier.
user_id	INTEGER	No	—	FK -> USERS.ID	Applicant ID.
job_listing_id	INTEGER	No	—	FK -> JOB_LISTINGS.ID	Target job listing ID.
status	VARCHAR	No	—	—	Application status lifecycle.
applied_date	DATETIME	No	datetime.now	—	Submission timestamp.
notes	TEXT	Yes	NULL	—	Candidate or internal notes.
created_at	DATETIME	No	datetime.now	—	Record creation timestamp.
updated_at	DATETIME	No	datetime.now	—	Last modification timestamp.

Constraints & Relationships:

- **Unique Constraint:** `uq_user_job_listing (user_id, job_listing_id)` ensures a user cannot submit duplicate applications for the same job.
- **Foreign Key Constraints:** `user_id` (**ON DELETE CASCADE**) and `job_listing_id` (**ON DELETE CASCADE**).
- **user / job_listing:** Many-to-One (**N:1**) relationship back to parent records.

3. Key Integrity Rules & Cascade Behaviors

Deletion Cascade Rules

- 1. Company Deletion:** Deleting a record in **companies** triggers database-level **ON DELETE CASCADE** on **job_listings**, followed by ORM-level cascade to delete associated **job_applications**.
- 2. User Deletion:** Deleting a record in **users** automatically purges all corresponding records in **job_applications** via foreign key constraint.
- 3. Job Listing Deletion:** Removing a **job_listing** automatically cleans up all submitted applications for that position.

Duplicate Application Safeguard

The explicit composite unique constraint **uq_user_job_listing** guarantees idempotency. Any database attempt to insert a duplicate **(user_id, job_listing_id)** tuple will raise an **IntegrityError**.